

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
RAM problem	Computer beeps and no display on the monitor
Power Supply (PSU) failure	PC does not turn on at all
Hard Disk (HDD/SSD) failure	Computer is very slow or Windows does not boot
CPU overheating	PC shuts down or restarts automatically
Graphics Card (GPU) problem	Blank screen, display flickers, or no display

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step	What to Check
1. Check the power outlet	Make sure the wall socket is working by plugging in another device.
2. Check the charger	Look for any damage to the charger or cable.
3. Check the charging port	See if the laptop charging port is loose, dirty, or damaged.
4. Check the battery	If the battery is removable, remove it and check for damage or swelling.
5. Use a multimeter	Measure the charger's output voltage to see if it is giving the correct power.
6. Try another charger	Use a compatible charger to check if the original charger is faulty.
7. Check the charging light	See if the charging LED turns on when the charger is connected.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Test	Measured Voltage	Expected Range	Result
USB Phone Charger Output	5.02 V	4.75 V – 5.25 V	Pass – Voltage is within the expected range.

4.A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Testing the 12V Rail of an ATX Power Supply

1. Turn off the PC and disconnect the ATX power connector from the motherboard.
2. Set the multimeter to DC Voltage (20V range).
3. Turn on the power supply (using a PSU tester or jumper if required).
4. Place the red probe on the yellow wire (12V).
5. Place the black probe on any black wire (Ground).
6. Read and record the voltage shown on the multimeter.

Test	Measured Voltage	Result
12V Rail (Yellow to Black)	12.08 V	Normal – Voltage is within the expected range.

If the voltage is below 11.5V:

It may indicate a faulty or failing power supply (PSU), which can cause the PC to randomly shut down or become unstable.